HYDROFARMS AGRI SOLUTIONS

NFT NUTRIENT TANKS, PUMPS, CHILLERS, PRESSURE FEED AND GRAVITY RETURN

Activity-Specific Construction Method Statement

Document field	Controlled value
Document code	HF-GH01-MST-NFT-001
Revision	00 - Draft for coordination
Project	One-Feddan NFT Lettuce Greenhouse Project
Location	Alexandria, Egypt
Hydraulic zones	East production, west production and nursery - fully independent
Prepared by	Hydrofarms Agri Solutions
Issue date	15 August 2026
Parent methods	HF-GH01-MST-CIV-003 and CIV-004

NO HYDRAULIC CROSS-CONNECTION
East production, west production and nursery solution circuits remain independent from tank outlet to tank return. The common standby production pump may be valved to either production circuit but shall never connect or mix the two tanks.

Document Control

Approval and responsibility

Role	Primary responsibility
Project manager	Resources, programme, subcontractor control and procurement status.
Construction manager	Work-front release, buried-services sequence and trade coordination.
Hydraulic engineer	Route, sizing, valves, testing, balancing and operating philosophy.
Refrigeration engineer	Chiller duty, refrigerant piping, SS316L coils, controls and tests.
Fertigation specialist	A/B/acid interface, EC/pH dosing and tank injection boundaries.
Surveyor	Route coordinates, inverts, slopes, table positions and as-built survey.
Electrical/controls engineer	Motor protection, level/flow proof, temperature control and alarms.
QA/QC engineer	Material traceability, pressure/gravity tests, flushing and turnover records.
HSE manager	Excavation, chambers, chemicals, pressure, refrigeration and electrical safety.
Client/consultant	Hold/witness inspection and technical acceptance.

Revision history

Rev.	Date	Purpose	Description
00	15 Aug 2026	Draft for coordination	Initial project-specific English issue

1. Purpose

This method controls installation, testing, flushing, cooling, balancing and handover of the treated-water reserve group and transfer pumps, nutrient operating tanks, NFT pumps, direct-expansion chillers with removable SS316L coils, buried pressure-feed mains, table feed assemblies, table drainage and buried gravity-return networks. It provides a continuous, cleanable and traceable path from treated-water storage to each plant channel and back to the correct operating tank.

2. Scope and Package Boundary

- Four 10 m3 treated-water reserve tanks, equalizing header, transfer pumps and sequential operating-tank fill connections.
- Two 10 m3 production nutrient tanks and one 3 m3 nursery tank, insulation, fittings, level/temperature instrumentation and local pump skids.
- Five 7.5 TR refrigeration units and five removable SS316L direct-expansion immersion coils.
- Three independent 3 in PN16 buried feed networks, table risers/manifolds and two 5/16 in feed points per NFT channel.
- Table drainage, 4 in SN8 return submains, 6 in production return mains, 4 in nursery return and tank-entry assemblies.

PACKAGE EXCLUSIONS
Groundwater well and WTP are separate. Concrete rooms, waterproofing and sumps belong to civil works. NFT tables/channels and their chassis belong to the table package, although their hydraulic interfaces are installed here. The fertigation skid and A/B/acid stock tanks are separate. Electrical feeders and control panels are measured in electrical/automation. Cooling-pad water is not part of this nutrient circuit.

3. Governing Information and Installation Basis

Priority	Controlled information
1	Approved AFC civil, hydraulic, refrigeration, electrical and control drawings
2	Latest CAD routes, table coding, chamber releases and tank/pump/chiller shop drawings
3	Supplier pump curves, chiller capacity data at 45 degC ambient and coil calculations
4	Manufacturer-approved uPVC pipe, fittings, solvent system and installation instructions
5	Approved fertigation interface and operating water-quality/temperature requirements
6	This method, ITP, test packs, cause-and-effect and commissioning plan

4. Hydraulic Zoning and Planting Load

Circuit	Tables	NFT channels	Operating tank	Hydraulic rule
East production	42	420	10 m3 east tank	East feed and return only
West production	36	360	10 m3 west tank	West feed and return only
Nursery	6	102 (17/table)	3 m3 nursery tank	Independent from production
Total	84	882	23 m3 operating volume	No circuit crossover

Every 16 m NFT channel receives nutrient solution at the feed cap and at the 8.00 m midpoint. Production tables carry ten channels; nursery tables carry seventeen channels. Returns flow by gravity from each channel to the same tank that supplied it.

5. Reserve and Operating Tank Systems

5.1 Tank and chamber schedule

System	Tank arrangement	Clear room dimensions	Key construction/interface
Reserve storage	4 x 10 m3 opaque food-grade HDPE/PE	12.80 x 3.70 x 3.00 m	One row; unchilled; common open-top room
Production operation	2 x 10 m3 insulated tanks	7.80 x 4.05 x 3.10 m	Shared room with internal divider
Nursery operation	1 x 3 m3 insulated tank	3.10 x 3.35 x 2.10 m	Independent room aligned with nursery

- Insulate each nutrient operating tank with 50 mm closed-cell insulation and weather/wash-resistant GRP cladding. Reserve tanks remain unchilled and uninsulated.
- Provide a lockable top inspection opening not less than 600 mm. Fit a 3 in vent to every 10 m3 tank and a 2 in vent to the 3 m3 nursery tank, protected against insects and rain.
- No permanent operating-tank overflow and no dedicated bottom washout are installed under the adopted design. High-high level closes the associated fill valve and stops transfer.
- Provide room floor fall to the civil sump, accessible isolation/union connections, pump/coil lifting path and removable 1 ton manual chain block/beam arrangement.

5.2 Treated-water reserve and transfer path

Line	Diameter	Centreline	Procurement	Function
WTP to reserve group	2 in PN16	45 m	50 m	Treated-water inlet, flow and EC/TDS points
Four reserve top-fill branches	1.5 in	12 m	14 m	Valve and flexible connector per tank
Reserve equalizing header	4 in	15 m	17 m	Low-level header; valve/flexible connector per tank
Transfer-pump suction	4 in	8 m	9 m	Flooded common suction; reduced to 3 in per pump
Transfer discharge	2 in	60 m	66 m	2 in to each production tank; 1.5 in to nursery

The four reserve tanks act as one controlled storage bank through the 4 in low-level equalizing header. Each tank remains individually isolatable. Two transfer pumps fill one operating tank at a time under level control; the standby pump alternates automatically.

6. Pumps and Standard Skid Connections

Service	Quantity/arrangement	Duty point	Power	Suction / discharge
Reserve transfer	2; duty/standby	12 m3/h at 25 m TDH	2.2 kW / about 3 HP	4 in header to 3 in / 2 in
Production NFT	3; 2 duty + common standby	30 m3/h at 30 m TDH	5.5 kW / about 7.5 HP	4 in / 3 in
Nursery NFT	2; duty/standby	8 m3/h at 25 m TDH	1.5 kW / about 2 HP	4 in / 3 in

6.1 Standard connection set for every pump

- Suction: full-bore isolation valve, large basket/Y-strainer, flexible connector, dismantling joint/union, vacuum-pressure gauge and eccentric reducer with flat side up.
- Discharge: silent non-return valve, full-bore isolation, dismantling joint/union, 0-6 bar pressure gauge, flow meter/indicator, sample point and 1 in bypass back to the associated tank where approved.
- Mount on a level 150 mm housekeeping plinth with anti-vibration pads. Keep flooded suction short, straight and free of high points.
- Provide low-level dry-run protection, motor overload/phase protection, lockable local isolator and Manual-Off-Auto selection.

The common production standby pump is connected through a fully valved selector manifold. Before changeover, isolate the inactive suction and discharge paths, prove valve position and confirm that the two production tanks cannot communicate through the standby skid.

7. Nutrient Chillers and Immersion Coils

Code	Served tank	Installed capacity	Refrigeration arrangement	Heat exchanger
CH-E-1/2	East production 10 m3	2 x 7.5 TR	Two independent Scroll/TXV circuits	Two removable SS316L coils
CH-W-1/2	West production 10 m3	2 x 7.5 TR	Two independent Scroll/TXV circuits	Two removable SS316L coils
CH-N-1	Nursery 3 m3	1 x 7.5 TR	Independent Scroll/TXV circuit	One removable SS316L coil
Total	Three operating tanks	5 x 7.5 TR = 37.5 TR	Five packaged air-cooled units	Five top-removable coils

- Chiller net capacity shall be guaranteed at 45 degC outdoor ambient. Units are tropicalized, air cooled and installed behind the fan side under small open-sided weather shades with full condenser/service clearance.
- Provide R-410A refrigeration package with Scroll compressor, TXV, filter-drier, sight glass, liquid solenoid, receiver, suction accumulator, HP/LP protection and anti-short-cycle timer.
- Route exposed copper on galvanized supports. Insulate and UV-jacket the suction line; keep detachable refrigerant joints above tank liquid level.
- Fabricate immersion coils and wetted supports in SS316L, independently removable from the top by the maintenance lifting beam without entering the tank.
- Use two PT100 sensors in SS316L pockets per tank: one for control and one independent verification/alarm. Target 20 degC; normal band 19-21 degC; high alarm 23 degC; critical 25 degC; local low-temperature cut-out 17 degC.
- Enable refrigeration only with safe tank level and confirmed NFT-pump operation. Existing NFT circulation mixes the solution; no separate tank-mixing pump is installed.

8. Buried Pressure-Feed Networks

Circuit	Pipe	Adopted centreline route	Calculated length	Procurement basis
East production	3 in uPVC PN16	28.2633 m main + 4 x 61.133 m branches	272.795 m	301 m incl. 10%
West production	3 in uPVC PN16	28.2633 + 2 x 43.0575 + 2 x 61.133 m	236.644 m	261 m incl. 10%
Nursery	3 in uPVC PN16	1.44 + 55.3059 + 8.037 + 16.0899 m	80.873 m	89 m incl. 10%
Total	3 in uPVC PN16	Three independent systems	590.312 m	651 m rounded

- Lay on accepted bedding with surveyed coordinates/depth and restrained fittings at changes. Use one compatible pipe/fitting/solvent system and allow full cure before testing.
- At each table provide a 3 x 1 in reducing tee/saddle, accessible inspection box, 1 in isolation valve, 1 in union and 1 in PN16 riser to the table feed header.
- Provide isolation at pump discharge, principal branches and every table. Label every branch with its greenhouse-zone/table code.
- Hydrotest each circuit separately at 1.5 times working pressure or the lower approved component limit. No concealed pressure line may be backfilled or covered before test and as-built survey release.

9. Table Feed Assemblies

Item	Production total	Nursery total	Technical requirement
Table risers / 1 in side headers	78	6	uPVC PN16; approximately 150 mm below channel level at valve bank
3/4 in transverse manifolds	156	12	Two balanced manifolds per table: feed cap and midpoint
3/4 in valves + unions	156 + 156	12 + 12	One isolation/union set per manifold
5/16 in microtubes	1,560 points / approx. 1,560 m	204 points / approx. 204 m	PE microtube; route without kinks
5/16 in mini ball valves	1,560	204	One individually adjustable valve per feed point
Barb + EPDM grommet entries	1,560	204	Leak-free compatible entry into channel
Flush/end closures	156	12	Removable end cap or flush valve per manifold

- Feed point A enters the approved channel feed cap. Feed point B enters at 8.00 m from the feed end. Both points remain within the channel and clear of planting openings.
- Align the valve bank below channel level, approximately 150 mm as approved, with each valve accessible and tagged by channel/plant coding.
- Flush each riser and manifold before connecting microtubes. Balance cap and midpoint flow, then verify uniform channel inlet flow without jetting or overflow.

10. Table Drainage Assemblies

Item	Production	Nursery	Requirement
1 in channel drain branches/elbows	780 + 780	102 + 102	One branch and elbow/drop per channel
3 in x 1 in 45-degree branches	780	102	Connect channel branches to table collector
3 in table collectors	78 / 187.2 m	6 / 14.4 m	Visible uPVC PN10; 1% fall
3 in vertical downpipes	78 / 70.2 m	6 / 5.4 m	Supported and aligned to buried return
3 in removable cleanouts	78	6	Accessible at collector end

Each NFT drain cap discharges through the dedicated 1 in branch to the sloped 3 in collector. The collector discharges through one 3 in downpipe. No channel may drain to the floor, another table or the wrong hydraulic zone.

11. Buried Gravity-Return Networks

Circuit	6 in main	4 in branches/submain	Centreline total	Procurement basis
East production	39.1911 m	2 x 61.133 = 122.266 m	161.457 m	178 m incl. 10%
West production	39.1911 m	43.335 + 61.133 = 104.468 m	143.659 m	159 m incl. 10%
Nursery	Not applicable	4.3 + 37.9 + 16.9 + 16.9 = 76 m	76.000 m	84 m incl. 10%
Total by diameter	78.382 m	302.734 m	381.116 m	6 in: 87 m; 4 in: 334 m

- Use heavy-duty SN8 uPVC gravity drainage with sealed sockets. Maintain 0.5% minimum and 0.7% preferred fall for production, and 1% for nursery, without reverse grade or trapped low points.
- Connect each table downpipe to the 4 in submain through a 4 x 3 in 45-degree wye and accessible 3 in cleanout cap.
- Provide inspection/cleanout access at every change of direction, collection point and at maximum 20-25 m intervals.
- Before each operating tank install a root/debris trap chamber with removable 1/8 in basket screen. Enter production tanks through 6 in returns and the nursery tank through a 4 in return, each with an internal downward elbow.
- Water-test by controlled filling and flow. Survey every invert before backfill and demonstrate unrestricted return to the correct tank under full table discharge.

12. Installation Sequence

- Release chambers, tank bases, buried routes, sleeves and table coordinates; confirm no conflict with foundations, floors, drains or electrical earthing.
- Install reserve and operating tanks, equalizing/fill headers, vents and instrumentation while lifting access remains open.
- Install pump skids, selectors, valves, gauges and temporary flushing connections; do not connect clean tanks to unflushed buried networks.
- Lay, survey, solvent-weld and test buried feed/return networks before floor closure. Protect open pipe ends at every shift.
- Install table collectors/downpipes and feed risers/manifolds after table alignment; then connect clean microtubes and channel drains.
- Install chillers, copper lines and removable SS316L coils; pressure/leak-test and evacuate refrigeration circuits before charging.
- Flush/disinfect water-contact systems, fill each circuit independently, balance table feeds, prove returns and commission temperature control.

13. Testing and Commissioning

Test	Method	Acceptance
Pressure feed	Hydrotest each zone separately	No pressure loss/leak at approved test pressure
Gravity return	Full-flow water test and invert survey	Continuous fall; no ponding, surcharge or wrong-tank return
Pump duty	Measure flow, pressure, current and vibration	Duty point within accepted pump curve/tolerance
Table feed balance	Measure cap/midpoint flow on representative and extreme tables	Uniform stable flow; individual valves accessible
Tank controls	Simulate low, high and high-high levels	Correct fill, stop, alarm and dry-run logic
Chillers	Run circuits at design load and record temperatures/pressures	19-21 degC control without icing or short cycling
Integrated run	Minimum 8 h with all three hydraulic zones	Stable EC/pH/temperature/level; no leakage or crossover

14. Inspection and Test Plan

No.	Inspection stage	Method	Type	Record
01	Shop drawings, equipment data and route schedule	Document review	H	Design release
02	Chambers, tank bases, sleeves and buried routes	Survey/visual	H	Civil/interface release
03	Tanks, pumps, chillers, coils and pipe materials	Identity/certificates	W	Material release
04	Buried feed joints before covering	Hydrotest/survey	H	Pressure test/as-built
05	Buried return inverts before covering	Water test/survey	H	Gravity test/as-built
06	Pump skids and selector valves	Mechanical/functional	H	Pump release
07	Table feed and drainage assemblies	Visual/flow test	W	Table hydraulic report
08	Refrigeration pressure/leak/vacuum	Instrumented test	H	Refrigeration test pack
09	Integrated hydraulic and temperature run	Performance test	H	Commissioning report
10	Training, spares, O&M and as-builts	Record review	R	Turnover dossier

Intervention codes: H = Hold, W = Witness, S = Surveillance, R = Record review.

15. Hold Points

- HP-NFT-01: Tank/pump/chiller selections, routes and hydraulic zoning accepted before procurement.
- HP-NFT-02: Chambers, bases, sleeves and lifting access accepted before tank/equipment installation.
- HP-NFT-03: Buried feed pressure tests and return invert/water tests accepted before covering.

- HP-NFT-04: Pump selector logic, valves and no-cross-connection proof accepted before filling nutrients.
- HP-NFT-05: Table feed/drainage balance and correct-tank returns accepted before planting.
- HP-NFT-06: Refrigeration leak/vacuum tests and SS316L coil installation accepted before charging.
- HP-NFT-07: Eight-hour integrated run, alarms, training and records accepted before handover.

16. HSE and Environmental Controls

Hazard	Mandatory control	Residual risk
Excavation/buried work	Permit, shoring, access, service scan and controlled backfill	Medium
Open tank rooms	Edge protection, safe ladder, ventilation, lighting and rescue plan	Medium
Pressure testing	Rated equipment, calibrated gauge, exclusion zone and controlled depressurization	Medium
Refrigerant work	Certified technician, ventilation, leak control, PPE and recovery equipment	Medium
Electrical/wet area	IP-rated equipment, earthing, LOTO and dry commissioning practice	Medium
Chemical/disinfection	Approved SDS, dosing procedure, PPE and controlled disposal	Medium
Confined access	No tank entry under this method; separate permit/rescue if ever required	Low
Wastewater	Discharge only to approved route; never to nutrient/reserve tanks or soil	Low

17. Records, Spares and Handover

- Approved layouts, hydraulic calculations, route/valve schedules, pump/chiller curves, coil calculations and tank nozzle drawings.
- Material certificates, pipe/fitting batches, solvent records, tank/insulation inspections and equipment nameplates.
- Pressure tests, return-invert surveys, water tests, flushing/disinfection and table balancing records by hydraulic zone.
- Pump duty, motor current, selector-valve proof, level alarms, flow proof and no-cross-connection tests.
- Refrigeration pressure/leak/vacuum/charge records, PT100 calibration and temperature-performance logs.
- As-built coordinates/inverts, O&M manuals, warranties, operator training and critical-spares register.